Comparative Analysis of Priming Effects across Different Drought Stress Durations in Basil (*Ocimum basilicum*)

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Abstract Recent climate change has hindered plant development, underscoring the need for strategies to strengthen plant tolerance against stress. Basil (*Ocimum basilicum*) is an herb vulnerable to water deficit, making it a suitable model for research on enhancing tolerance through drought priming. This study observed the priming effect in basil induced by drought stress and examined how it differs with stress duration. A control group without primary drought stress and five experimental groups (G0 to G4), each given a primary drought stress whose duration differed in 2-day steps, were established. After recovery, all groups received the same secondary drought stress. Using the RGB color code as an indicator, the rate of change in G-ratio was analyzed and compared with the change in plant angle. A priming effect was observed in the experimental groups conditioned to drought stress. Under this study's conditions, the priming effect was most evident in the group with a 10-day stress period. This result suggests that, by regulating the stress duration, priming can be used as a strategy for enhancing basil's stress tolerance.

Key Words: Basil (*Ocimum basilicum*), Drought Stress, Priming

I. Introduction

Plant priming refers to a phenomenon in which a plant, after experiencing an initial stress stimulus, acquires resistance and shows a faster and stronger defense response to the same or similar stress later. As accelerating global climate change has made declining agricultural productivity an important issue, priming, which is directly linked to plant stress tolerance, has been studied continuously as a promising approach for improving plants' defensive capacity and survival.

The basic mechanism of priming is based on the "stress memory" a plant forms when it is first exposed to stress. Stress memory helps the plant adapt to stress by applying antioxidant, osmotic, hormonal, and ROS regulation, reducing growth inhibition [1]. Priming falls into three types: seed priming, which induces tolerance at the germination stage; foliar priming, which applies priming agents such as chemicals directly to the leaves; and root priming, which exposes the roots to abiotic stress [2]. This suggests that priming effects can be applied across all growth stages, from seed to mature plant.

Stresses that induce priming include not only biotic stress but also abiotic stresses such as chemical stimuli, drought, temperature, and salt [3]. Among these, drought stress is the one plants are most likely to face during growth, and it is directly tied to survival. In particular, unstable rainfall caused by climate change and various agricultural conditions often leave plants unavoidably exposed to drought stress.

However, previous research has focused on seed priming or chemical-based priming. Natural stressors such as drought involve relatively many confounding variables that can affect experimental results, so they have been studied less actively.

Several earlier studies report that drought-induced priming differs by plant species. In the ornamental flower petunia (*Petunia* × *hybrida*), a priming effect that enhanced tolerance was observed under drought stress [4], while in the argan tree (*Argania spinosa*), which is relatively insensitive to drought because of its native

habitat, no priming effect appeared [5]. These contrasting results show that drought-induced priming varies by species and that crop-specific research is needed. Basil (*Ocimum basilicum*), chosen for this study, belongs to a family of herbs sensitive to drought stress [6], so we judged it a suitable model for determining whether a priming effect occurs.

This study began from a priming-like phenomenon observed under uncontrolled, natural conditions while growing basil. Basil that was rewatered after a period of drought stress recovered within 72 hours, and when watering was later restricted again, it survived longer. This suggested that drought stress caused a priming effect that improved basil's survival, and it raised the need for a study under controlled conditions.

This study aimed to test experimentally whether the duration of drought stress affects the degree and pattern of the priming effect. Because a moderate level of drought stress induces priming, but longer periods may cause irreversible physiological damage, we also aimed to investigate the maximum tolerable duration of drought stress.

II. Methodology

1. Research Design

This study was designed to determine how priming affects basil's stress tolerance under drought. We set the hypothesis that "when the duration of the primary drought stress applied to basil is within a certain range, the priming effect will increase in proportion to that duration," and designed the experiment accordingly. Basil seedlings were divided into a control group and five experimental groups, each given a different primary drought stress duration. After a recovery phase, a secondary drought stress was applied to the control group and all experimental groups. Through this design, we analyzed how the presence and duration of priming affect the plant's response to later drought stress.

The independent variable was the duration of the primary stress, and the dependent variables were the color of the stem and leaves and the drooping angle of the leaves under drought stress. To analyze the relationship between the two systematically, the same environmental conditions were applied to every group to minimize the influence of factors other than drought stress. The experiment followed a completely randomized design (CRD) with drought stress duration as the single factor, and nine basil plants were randomly assigned to each group.

2. Materials

To analyze how primary stress affects basil's stress response and recovery, 54 basil seedlings in similar growth condition were used and planted individually in pots of the same size (6 × 4 × 5 cm). Pot size and shape were identical in all groups to rule out differences in water retention due to soil volume.

Regular potting soil and dried potting soil were used. Regular soil was used for the control group and whenever plants needed to grow normally, and dried soil was used to apply fast and extreme drought.

The experiment was conducted indoors, with an LED plant-growth panel as the light source to keep lighting uniform. Because natural light varies with time of day and weather, the growing space was enclosed with cardboard boxes and blackout cloth to minimize outside light.

Because the amount of water directly affects the results, every plant received the same amount of water (50 mL).

3. Procedure

The 54 seedlings were divided into six groups (Control, G0, G1, G2, G3, G4) of nine plants each. Because each experimental group received a primary drought stress of different length, the schedule was arranged accordingly. The primary stress, recovery, and secondary stress periods of each group are shown in Table 1. "Day 1" is defined as the day the primary stress began for all experimental groups, and all dates are counted from it. The control group was put under drought stress starting on Day 14, the day G2's secondary stress began. Following the hypothesis that a longer primary stress strengthens priming, the control group's stress was started at the same time as G2, the group with the median primary stress length, so that the control and experimental groups could be compared fairly.

Group	Day 1 to Day 22																					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Control														D	D	D	D	D	D	D	D	D
G0	D	D	D	D	D	D	R	R	R	D	D	D	D	D	D	D	D	D	D	D	D	D
G1	D	D	D	D	D	D	D	D	R	R	R	D	D	D	D	D	D	D	D	D	D	D
G2	D	D	D	D	D	D	D	D	D	D	R	R	R	D	D	D	D	D	D	D	D	D
G3	D	D	D	D	D	D	D	D	D	D	D	D	R	R	R	D	D	D	D	D	D	D
G4	D	D	D	D	D	D	D	D	D	D	D	D	D	D	R	R	R	D	D	D	D	D

Drought Recovery (watered every 24 h) Normal growth

Table 1. Stress schedule of each group. Control: 9-day drought. G0: 6-day drought, 3-day recovery, 13-day drought. G1: 8 / 3 / 11. G2: 10 / 3 / 9. G3: 12 / 3 / 7. G4: 14 / 3 / 5.

Drought method

Drought stress is created by restricting the plant's water supply. To create a uniform and intense drought environment in a short time, this experiment used potting soil that had been fully dried in advance. The soil was air-dried outdoors for three days before the experiment to remove as much internal moisture as possible, then packed into pots of the same size. Basil growing in regular soil was carefully pulled out to minimize root damage, and the soil attached to the roots was lightly shaken off. The roots were not washed, to avoid physical damage to root tissue. The basil was then immediately transplanted into the pots filled with dried soil. This sharply lowered the soil moisture of the whole pot and created a strong drought environment within a short period. This method reduces variation between conditions and allows clear control over when the stress begins.

Primary drought stress

The primary drought stress was applied for priming. The experimental groups were designed so that basil experienced a temporary water deficit for a set period, and each group received a different duration so that responses could be compared by stress intensity. Watering was fully stopped and the dry state was maintained for the set number of days, letting soil moisture deplete gradually. During this period, external changes such as leaf drooping, color change in the leaves and stem, and slowed growth were observed.

Recovery

After the first drought stress, all experimental groups were watered and given a three-day recovery period. During recovery, water was supplied every 24 hours to keep the soil from drying and to maintain normal growing conditions. This phase separates the primary and secondary stresses and is essential for judging the priming effect.

Secondary drought stress

The same secondary drought stress was applied to the control group and all experimental groups. It was used to evaluate the presence and relative strength of priming by comparing responses according to whether a plant had experienced the primary stress. Drying was carried out in the same way as the primary stress, and the condition of each group was compared by drought day to derive the results.

4. Measurement

To measure basil's physiological changes under drought stress, three image-based indicators were used: stem color analysis, leaf color analysis, and plant angle analysis.

Photos were taken of every group during the secondary drought stress. All photos were taken under the same white LED lighting, every 24 hours at the same time of day (3 p.m.). To reduce error in angle measurement, the camera height was set so that the soil surface was level with the bottom of the lens, and the lens was kept 10 cm from the plant. Each plant was photographed from the front and the side, for color analysis and angle analysis respectively. Every image included the whole stem and all leaves.

Stem color analysis

The RGB indicator, which allows color to be quantified from digital images, was used. From the front images, the color of every plant's stem was sampled 1 cm above the soil surface, and the RGB values (R, G, B; $0 \leq R, G, B \leq 255$) were recorded.

Previous research found that the proportion of green (G) in a plant's RGB color is positively correlated with total chlorophyll content [7]. As drought stress continues, chlorophyll-degrading enzymes increase and break down chlorophyll molecules, lowering the G-ratio. The G-ratio was therefore defined as the measure of drought stress and used for the stem color analysis.

$$\text{G-ratio} = G / (R + G + B)$$

The mean G-ratio of each group was then calculated in R and plotted for comparison.

Leaf color analysis

Leaf color was analyzed in the same way as the stem, using the RGB indicator and calculating each plant's G-ratio. The color was sampled at the exact center of the leaf midrib.

Plant angle analysis

Plant angle was analyzed to supplement the color indicators. To evaluate wilting, a representative plant showing the average condition of each group was selected and its angle was analyzed by day. Side images were used, and three reference points A, B, and C were defined to obtain ∠ABC (in degrees).

A: the point where a perpendicular dropped from B meets the soil surface.
B: the point where the stem bends the most.
C: the point where the leaf begins.



Figure 1. Reference points A, B, and C set for plant angle measurement (left: Group 4, Day 18; right: Group 4, Day 19)

∠ABC was calculated using the points defined above. This method minimized angle errors caused by differences in stem length or plant shape.



Figure 2. Calculating ∠ABC with a protractor program (left: Group 4, Day 18; right: Group 4, Day 19)

III. Results

The experiment used a control group and five experimental groups that received primary drought stress of different lengths. Using the methods above, quantitative data were collected for every plant: stem G-ratio, leaf G-ratio, and plant angle. The G-ratio reflects changes in color and chlorophyll content, so its graphs describe the physiological effects of drought stress. Leaf angle focuses on plant form and describes changes

in the plant's support structure under drought stress. This study drew its conclusions from these quantified stress values.

The four graphs below were produced in R. In Figures 3, 4, and 5, the x-axis is the number of days since G0's primary stress began (as in Table 1), and the y-axis is the mean G-ratio or leaf angle.

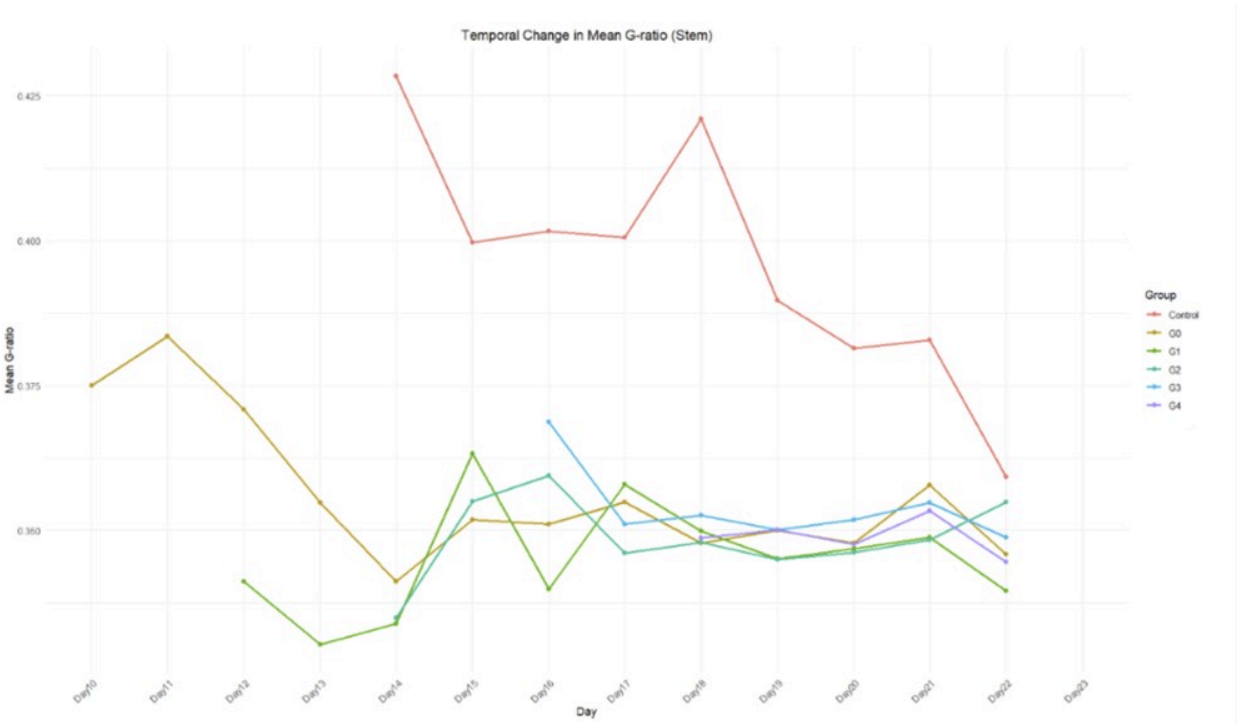


Figure 3. Change in mean stem G-ratio by day

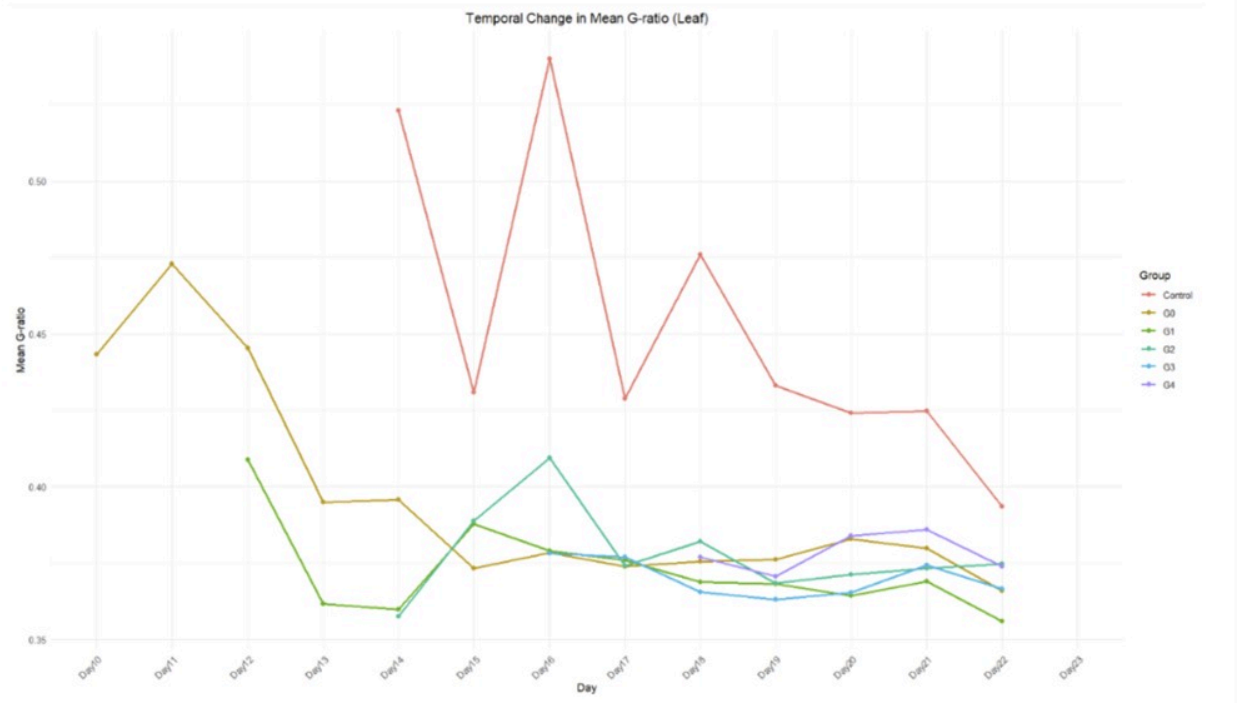


Figure 4. Change in mean leaf G-ratio by day

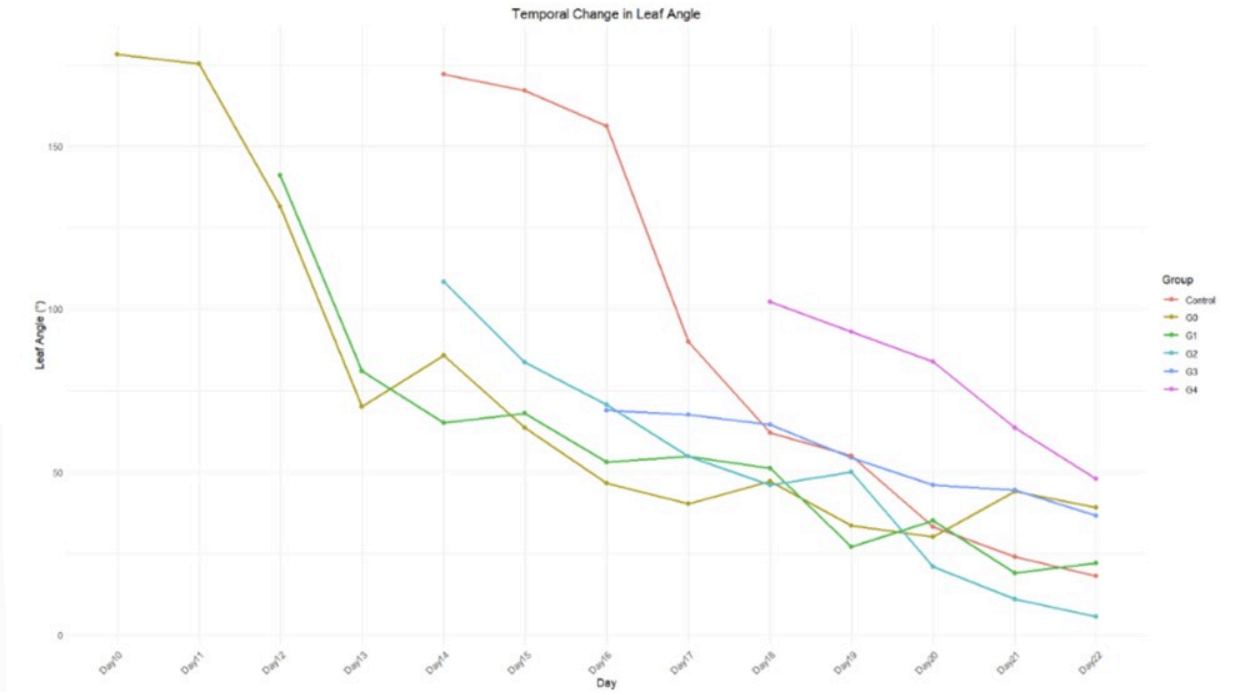


Figure 5. Change in leaf angle by day

In Figure 6, the x-axis is the number of days since G0's primary stress began, the left y-axis is the mean G-ratio, and the right y-axis is the leaf angle.

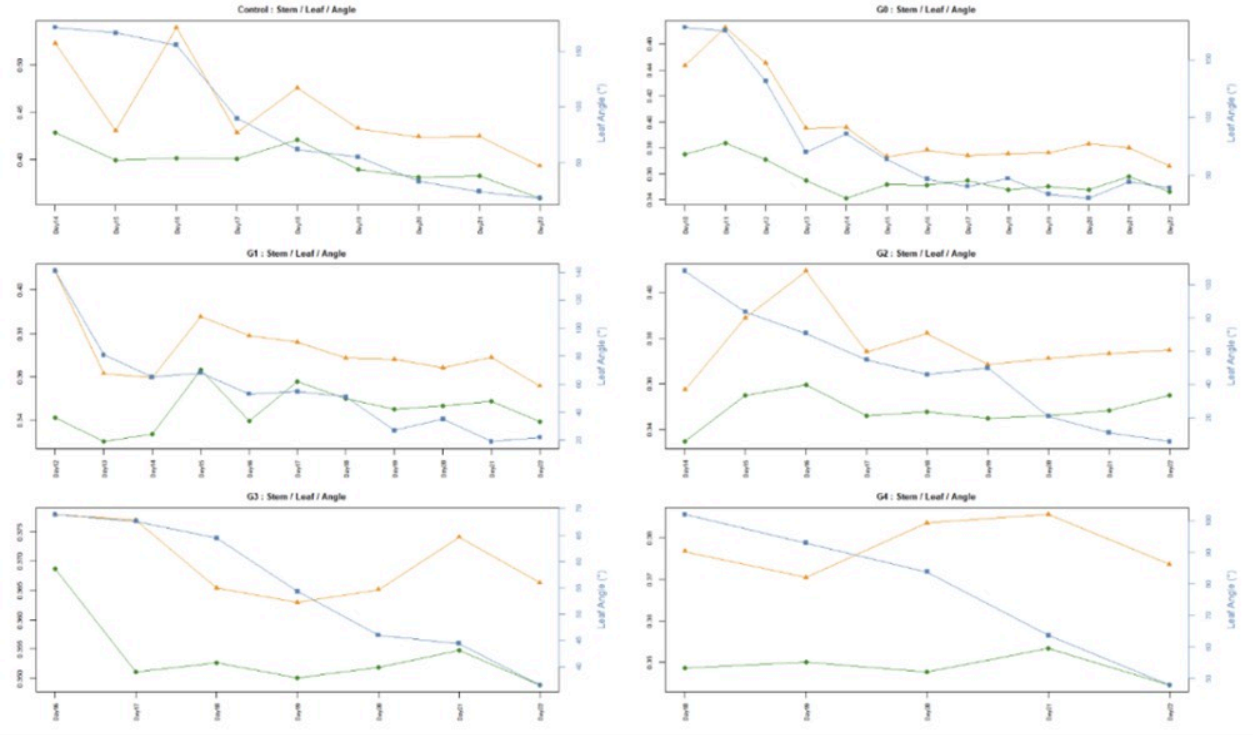


Figure 6. Mean stem and leaf G-ratio and leaf angle of each group over the experiment (orange: mean leaf G-ratio; green: mean stem G-ratio; blue: leaf angle)

Figure 3 shows the change in mean stem G-ratio for each group. The control group shows an overall decline over time. G0 has the highest initial value of the five experimental groups and declines sharply from Day 11

to Day 14. G0's high initial value is presumed to come from its having the shortest primary stress. From Day 18 to Day 22, G0 to G4 rise and fall over almost the same intervals and show similar trends.

Figure 4 shows the change in mean leaf G-ratio for each group. The control group generally declines but spikes briefly on Day 16. As in Figure 3, G0 has the highest initial value of the five experimental groups and declines steadily until Day 17. From Day 18 to Day 22, G0 to G4 rise and fall over similar intervals, and their trends largely match.

Figure 5 shows the change in leaf angle for each group. The control group and all experimental groups show an overall decline as the secondary stress continues. G0 and G1 drop sharply on Day 13, and the control group on Day 17. On Day 22, G2 has the lowest value, G1 is similar to the control group, and G0 is similar to G3. The group with the highest angle on Day 22 is G4.

The trends in the G-ratio and leaf angle graphs carry their own meaning and are an important indicator alongside each group's data. If the stem G-ratio graph (Figure 3) and the leaf G-ratio graph (Figure 4) rise and fall together, chlorophyll breakdown is not limited to one organ but proceeds at a similar intensity throughout the plant. In fact, during most of the secondary stress, the stem and leaf G-ratio moved in almost the same direction in every group, including the control group, which implies that drought stress had a physiological effect on the whole plant. In some intervals, however, the stem and leaf G-ratio trends were opposite, which suggests that physiological effects differ by plant structure. In those cases, one organ receives relatively more stress than the others: if the stem G-ratio rises while the leaf G-ratio falls, the effect of drought is concentrated in the leaves, and in the opposite case it is concentrated in the stem.

The leaf angle trend can also be interpreted in terms of basil's structural and functional responses. Leaf angle responds sensitively to structural effects and can complement the G-ratio data when the latter is uncertain. Considering increases and decreases in both leaf angle and G-ratio, a plant's drought stress pattern can be divided into four cases: (1) angle up, G-ratio up; (2) angle up, G-ratio down; (3) angle down, G-ratio up; (4) angle down, G-ratio down. When both rise, both structural and physiological indicators are stable and the effect of drought during that period is minimal. When the angle rises while the G-ratio falls, the physiological indicator is less stable than the structural one, so the physiological stress load has increased. When the angle falls while the G-ratio rises, the plant is structurally unstable but the physiological effect is relatively small. When both fall, the effect of drought stress during that period was strong and systemic. This interpretation judges the data together; rather than analyzing each group separately, it analyzes the combined trend to obtain a result that represents all indicators.

Figure 6 was analyzed to check how similar the stem and leaf G-ratio trends are. In this graph, the stem and leaf G-ratio show similar trends in each group, so drought stress can be considered to have caused a systemic response in most plants in the experiment. A detailed analysis of each group in Figure 6 follows.

In the control group, the leaf angle decreased steadily over time, while the G-ratio alternated between increases and decreases from Day 14 to Day 18. From Day 15 to Day 16 and from Day 17 to Day 18, the G-ratio increased while the leaf angle decreased, which is interpreted as the structural stress response dominating the physiological indicator. Outside these intervals, decreases in leaf angle and G-ratio were observed together, indicating a systemic effect of drought stress.

In G0, both leaf angle and G-ratio fluctuated over time. From Day 10 to Day 11, the leaf angle decreased and the G-ratio increased, so structural stress dominated; from Day 11 to Day 13, both decreased together, showing a strong systemic response to drought. Afterward, intervals dominated by structural and by physiological responses alternated, but from Day 20 to Day 21 both leaf angle and G-ratio increased, which is interpreted as the effect of drought stress being relatively small during that period.

In G1, the leaf angle generally decreased while the G-ratio alternated between increases and decreases. From Day 12 to Day 14, both decreased together, indicating a large systemic response. From Day 14 to Day 15, the leaf angle decreased and the G-ratio increased, so the structural response dominated the physiological change. Afterward, the decrease in leaf angle and the fluctuation in G-ratio continued, showing that structural and physiological responses alternated.

In G2, the leaf angle generally decreased while the G-ratio alternated between increases and decreases. From Day 14 to Day 16, the leaf angle decreased while the G-ratio increased, showing that structural stress dominated. From Day 18 to Day 19, the leaf angle increased while the G-ratio decreased, meaning the physiological response to drought dominated. From Day 19 to Day 22, the leaf angle decreased while the G-ratio increased, suggesting that G2's response shifted to a structural one.

In G3, the leaf angle decreased steadily throughout the period. In most of the period except Day 19 to Day 21, leaf angle and G-ratio decreased together, so drought stress acted strongly. From Day 19 to Day 21, the leaf angle decreased and the G-ratio increased, which is interpreted as drought stress appearing mainly as a structural problem during that period.

In G4, the leaf angle generally kept decreasing while the G-ratio alternated between increases and decreases. From Day 18 to Day 19 and from Day 21 to Day 22, leaf angle and G-ratio decreased together, clearly showing a systemic effect. From Day 19 to Day 21, the leaf angle decreased and the G-ratio increased, which is interpreted as structural stress from drought being relatively dominant.

This analysis assumes that physiological phenomena such as chlorophyll breakdown and the plant's structural form are affected by drought stress at the same time. Because this assumption may not hold under all conditions, other cases can be considered. A difference in sensitivity between tissues is one case in which physiology and structure are not affected simultaneously. If tissues within one plant differ in sensitivity, this can explain differences between the G-ratio and angle trends: when drought stress is applied, the stem cells, leaf cells, and plant structure may respond at different times, with different intensities, and in different ways. In that case, the G-ratio and the angle may show different trends even under the same level of stress. A time lag is another possible effect of differing tissue sensitivity. If there is a time lag between angle and G-ratio, structural change is likely to occur more slowly than leaf color change, so the angle trend would appear later than the G-ratio trend. If such tissue-specific sensitivity or time lag exists, physiology and structure may not be affected to a similar degree by drought stress. Further verification of tissue-specific sensitivity or organ-level time lag is needed to address possibilities not covered by these results.

IV. Discussion

To analyze whether a priming effect appears in basil (*Ocimum basilicum*) depending on drought stress duration, 54 basil seedlings were divided into a control group and five experimental groups. Based on the hypothesis that "when the duration of the primary drought stress applied to basil is within a certain range, the priming effect will increase in proportion to that duration," each experimental group received a different primary drought stress duration (the control group received no primary stress). After the primary stress, each group was given three days of recovery and then immediately the secondary stress. During the secondary stress, the mean stem G-ratio, mean leaf G-ratio, and plant angle of each group were measured and summarized in Figures 3, 4, 5, and 6. Figure 6 compares the mean stem and leaf G-ratio and plant angle by group and is useful for comparing their trends.

Figure 6 shows that the stem and leaf G-ratio rise and fall over almost the same intervals. The G-ratio is determined by chlorophyll content, and under drought stress the amount of chlorophyll-degrading enzyme in

the plant increases and breaks down chlorophyll molecules. Given this, matching intervals for the leaf and stem G-ratio suggest that chlorophyll breakdown proceeds at a nearly similar level regardless of organ, showing that the physiological effect of drought stress occurs similarly across all organs. In addition, the stem and leaf G-ratio trends and the plant angle trend mostly decrease together. This can be interpreted as drought stress affecting the plant's structure as well as its physiology, that is, having a systemic effect. Without a systemic effect, the G-ratio and leaf angle would have shown opposite trends.

Figures 3, 4, and 5 show large differences in initial values between groups. The initial value is each group's value on the day its secondary stress began. The difference in primary stress duration is one of the largest factors behind these differences. However, Figure 3 shows that a longer primary stress does not always produce a lower initial value. The initial stem G-ratio was highest in the control group, then decreased in the order G0, G3, G4, G1, and was lowest in G2. Two hypotheses were formed to explain this. The first is that if the primary stress lasts beyond a certain period, the stem recovers through some mechanism. The second is that if drought stress lasts beyond a certain period, salts accumulate and a white powder forms on the stem surface. The G-ratio of the white powder is relatively higher than that of a stem browned by drought, so plants with the powder may show higher G-ratio values than expected. Further research and more precise analysis are needed to verify both hypotheses.

V. Conclusion

In this study, the priming effect by drought stress duration was determined by confirming priming through the slope (instantaneous rate of change) of the G-ratio graphs and by observing priming patterns through comparing the G-ratio with plant angle.

First, the presence of a priming effect was checked using the slope of the G-ratio graph. The slope represents the instantaneous rate of change of the G-ratio, and the larger its value, the larger the priming effect. When the slope is negative, a smaller absolute value means less change in total chlorophyll due to drought stress and thus a larger priming effect. When the slope is zero or positive, a larger absolute value means less effect from drought stress and a larger priming effect. It can therefore be concluded that the strength of the priming effect is proportional to the instantaneous rate of change of the G-ratio graph.

Figures 3 and 4 show that the control group, which received no primary drought stress, has a large negative slope. The experimental groups that went through primary drought stress (G0 to G4) decreased with a smaller absolute slope or even increased in some intervals. From this, it can be inferred that basil primed with drought stress became more tolerant of stress, and that priming strength increases with the duration of the primary drought stress.

The relationship between increases and decreases in G-ratio and angle from the results is summarized in Table 2.

G-ratio	Plant angle	Result
Decrease	Decrease	Systemic effect: structural instability, reduced physiological capacity
Decrease	Increase	Physiological effect: reduced physiological capacity, structural stability maintained
Increase	Decrease	Structural effect: structural instability, physiological capacity maintained
Increase	Increase	Relatively little effect from stress

Table 2. Relationship between G-ratio and plant angle

Based on this table, a systemic effect means that the plant's structural stability and physiological capacity declined at the same time, so of the four cases it represents the most damage from drought stress. To check

the degree of damage, the proportions of systemic, structural, and physiological effects in each group were summarized from Figure 6.

	Control	G0	G1	G2	G3	G4
Systemic effect	75%	58%	40%	13%	67%	50%
Structural effect	25%	25%	20%	75%	33%	50%
Physiological effect	0%	17%	30%	13%	0%	0%

Table 3. Proportion of each effect by group

The control group has a systemic effect proportion of 75%, the highest of all groups. This can be seen as the control group, which was not primed, suffering greater damage from drought stress. In contrast, the five experimental groups show systemic effect proportions of 58% (G0), 40% (G1), 13% (G2), 67% (G3), and 50% (G4), indicating that tolerance to drought stress developed, that is, that priming took place.

The strength of the priming effect can also be estimated from the proportion of systemic effect. In G0, G1, and G2, the proportion of systemic effect decreased as the primary stress period increased, so a longer primary stress was expected to produce a stronger priming effect. However, in G3 and G4 the proportion rose again. We attribute this to the possibility that, in G3 and G4, accumulated drought stress exceeded the plant's recovery limit and hindered the priming effect. We therefore judged that the priming effect was clearest in G2, the threshold point.

In conclusion, this study evaluated the presence and strength of the priming effect and the appropriate stress duration by comparing the groups. The instantaneous rate of change of the G-ratio confirmed that a priming effect was observed in basil. In addition, the trend analysis between G-ratio and plant angle showed that priming was most effective when the primary stress lasted 10 days. If the period exceeds 10 days, a priming effect may not appear. We therefore concluded that "the presence and strength of priming are determined in proportion to the primary stress period, but beyond 10 days the plant may pass a threshold and fail to recover."

These findings can be used to strengthen basil's stress tolerance. Because basil is a crop of high commercial value, the results are expected to have strong practical applications in agriculture. Future research could extend this approach by applying drought stress priming to other plants vulnerable to environmental stress.

VI. References

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